

Groundnut (Peanut/Mungphali) Farming Guide for Pakistan

A complete crop guide for farmers, students, and agriculture advisors

1. Introduction

Groundnut, known locally as Mungphali, is an important oilseed and cash crop grown in Pakistan, valued both for its edible nuts and its oil content. Unlike most crops, groundnut is unique because its flowers develop above the ground but the pods (containing seeds) mature underground, a feature that shapes many of its cultivation practices. Groundnut is popular as a snack food, especially roasted and eaten during winter, and is also used in the production of cooking oil, peanut butter, and confectionery products. It is grown mainly as a rain-fed (barani) crop in Pakistan, making it an important source of income for farmers in areas without reliable irrigation. The crop also improves soil fertility because, like other legumes, it fixes atmospheric nitrogen through root nodules.

2. Scientific Name

The scientific name of groundnut is *Arachis hypogaea*, belonging to the family Fabaceae (legume family). It is also commonly known as peanut, earthnut, or monkey nut in different parts of the world, while in Pakistan it is universally called Mungphali.

3. Importance in Pakistan

Groundnut is an economically important crop in Pakistan, providing both food and cash income for many rain-fed farming communities, especially in the Pothohar region of Punjab. As an oilseed crop, it contributes to the country's cooking oil supply, helping reduce reliance on imported edible oils. Groundnut is also valuable in crop rotation systems because it fixes nitrogen in the soil, improving fertility for the following crop, often wheat. The crop provides good returns per acre compared to some traditional rain-fed crops, and its by-products, including groundnut cake, are used as a protein-rich livestock feed. Roasted groundnuts are also a popular and nutritious snack food consumed widely across Pakistan, especially in winter.

4. Major Producing Provinces and Districts

Groundnut cultivation in Pakistan is concentrated mainly in specific rain-fed regions. - **Punjab**: The Pothohar plateau, including Chakwal, Attock, Jhelum, and Rawalpindi districts, is the leading groundnut-growing area in the country, mostly under rain-fed conditions. - **Khyber Pakhtunkhwa**: Grown in Kohat, Karak, and parts of Mardan and Swabi, generally on a smaller scale than Punjab. - **Sindh**: Limited groundnut cultivation exists in some irrigated areas, though it is much less significant compared to Punjab. - **Balochistan**: Groundnut is grown on a small scale in scattered locations with suitable soil and rainfall. Chakwal district in Punjab is often considered the heartland of groundnut production in Pakistan.

5. Climate Requirements

Groundnut is a warm-season crop requiring temperatures between 25°C and 30°C for optimal growth, with germination occurring best around 20°C or above. It needs a frost-free growing season of about four to five months and is sensitive to cold temperatures, especially during germination and pod development. The crop requires moderate rainfall of about 500 to 750 millimeters, well distributed throughout the growing season, with a dry spell preferred during the maturity and harvesting period to allow proper pod ripening and to reduce disease risk. Groundnut is generally grown as a rain-fed crop in Pakistan, relying on monsoon rains for most of its water needs.

6. Soil Requirements

Groundnut grows best in light-textured, well-drained sandy loam soils, since its pods develop underground and need loose soil for easy pegging (the process where flower stalks push into the soil) and pod formation. Heavy clay soils are not suitable because they restrict pod development and make harvesting difficult. The ideal soil pH range is between 6.0 and 7.5, with slightly acidic to neutral soils generally preferred. Good drainage is essential, as waterlogging can severely damage the crop, particularly during pod formation and maturity stages.

7. Land Preparation

Thorough land preparation is important for groundnut because of its unique underground pod development. The field should be ploughed deeply two to three times to loosen the soil, followed by harrowing and leveling to create a fine, crumbly seedbed free of clods, which allows easy pegging and pod formation. Ridges or raised beds are sometimes prepared, especially in areas with heavier soils, to improve drainage and aeration around the developing pods. Removing stones and large clods from the field surface is particularly important for groundnut, more so than for many other crops, since these can physically obstruct pod development.

8. Recommended Varieties (Pakistan)

Several improved groundnut varieties have been developed and recommended for cultivation in Pakistan. - **BARD-479**: A widely grown variety known for good yield and oil content. - **BARI-2000**: Recommended for the Pothohar region, offering good drought tolerance. - **Golden**: An older but still popular variety among Punjab farmers. - **Chakwal-2**: Suited specifically to the rain-fed conditions of the Chakwal region. - **BARI Groundnut-2011**: A more recent variety with improved yield potential. Farmers should obtain certified seed from provincial seed corporations or agricultural research institutes to ensure good germination and yield potential.

9. Seed Rate

The recommended seed rate for groundnut depends on the variety type (bunch or spreading type) and seed size. - For bunch-type varieties: 40 to 45 kilograms of shelled seed per acre. - For spreading-type varieties: 30 to 35 kilograms of shelled seed per acre, since these varieties need more space between plants. It is important to use fresh, good quality seed with high germination percentage, since old or poorly stored groundnut seed loses viability quickly.

10. Seed Treatment

Before sowing, groundnut seed should be treated with a fungicide such as Thiram or Captan to protect against seed-borne fungal diseases and improve seedling establishment. Since groundnut is a legume, seed can also be treated with Rhizobium bacterial inoculant, which helps improve nitrogen fixation in the root nodules, especially in fields where groundnut has not been grown recently. Seed should be treated shortly before sowing, handled gently to avoid damaging the seed coat, and treated seed should never be used for consumption.

11. Sowing Time (Province-wise)

- **Punjab:** In the Pothohar region, groundnut is typically sown from May to June, timed to make use of pre-monsoon and monsoon rains.
- **Khyber Pakhtunkhwa:** Sowing generally takes place from April to May in Kohat and Karak districts.
- **Sindh:** Where grown, sowing under irrigated conditions can take place from February to March.
- **Balochistan:** Sowing is usually done in April to May, depending on local climate. Sowing at the correct time ensures the crop benefits fully from monsoon rainfall during its critical flowering and pegging stages.

12. Sowing Methods

Groundnut is typically sown using the following methods. - **Ridge sowing:** Seeds are sown on ridges with furrows in between, which improves drainage and makes pegging and harvesting easier; commonly recommended for heavier soils. - **Flat sowing with row spacing:** Used mainly in lighter, well-drained sandy soils, with row spacing of 30 to 45 centimeters and plant-to-plant spacing of about 10 to 15 centimeters. - **Drilling with seed drill:** Increasingly used on larger farms for uniform seed placement and spacing. Proper spacing is important to allow enough room for lateral branches to peg into the soil and form pods.

13. Fertilizer Recommendations (NPK)

Being a legume, groundnut requires relatively less nitrogen than cereal crops, but responds well to phosphorus and potassium. - **Nitrogen (N):** 10 to 15 kilograms per acre, applied as a starter dose at sowing, since the crop fixes most of its nitrogen requirement through root nodules. - **Phosphorus (P2O5):** 30 to 40 kilograms per acre, applied fully at sowing time, as phosphorus is important for root development and nodulation. - **Potassium (K2O):** 20 to 25 kilograms per acre, important for pod filling and oil content. - **Gypsum (calcium sulfate):** Application of about 100 to 150 kilograms per acre at flowering time is often recommended, as calcium is particularly important for proper pod and kernel development in groundnut. Balanced fertilization, along with proper Rhizobium inoculation, helps maximize both pod yield and oil content.

14. Irrigation Schedule

Under irrigated conditions, groundnut requires about five to six irrigations spaced through the growing season, with critical stages being flowering, pegging, and pod development. The first irrigation should be given at sowing if soil moisture is insufficient, followed by irrigations roughly every 15 to 20 days depending on soil type and weather. Irrigation should be reduced or stopped as the crop approaches maturity, since excess moisture at this stage can cause pod rot and delay proper drying and harvesting. In rain-fed areas, no supplemental irrigation is applied, and crop success depends heavily on the timing and amount of monsoon rainfall.

15. Weed Management

Weed control is particularly important in groundnut because dense weed growth can interfere with pegging and pod formation. Effective management includes one or two hand weeding or hoeings at around 20 and 40 days after sowing, timed before the pegging process begins so that soil disturbance does not damage developing pegs later in the season. Pre-emergence herbicides can also be used in larger commercial fields, applied according to label instructions. Keeping fields weed-free during the first six to eight weeks is especially critical for good pod set and yield.

16. Insect Pests

Groundnut crops in Pakistan can be affected by several insect pests. - **Jassids (leafhoppers):** Suck sap from leaves, causing yellowing and reduced plant vigor, sometimes called "hopper burn." - **Thrips:** Damage young leaves and flowers, reducing photosynthesis and flower retention. - **Termites:** Can damage roots and developing pods, especially in dry, sandy soils. - **Aphids:** Occasionally infest young plants, particularly in dry weather. Regular field monitoring and use of recommended insecticides, applied only when pest populations exceed damaging levels, help protect yield while minimizing unnecessary chemical use.

17. Diseases

Several diseases can affect groundnut crops in Pakistan, particularly under humid conditions. - **Leaf spot diseases (early and late leaf spot):** Cause circular brown or dark spots on leaves, leading to premature leaf drop and reduced yield. - **Collar rot:** A fungal disease affecting the stem base, particularly in poorly drained soils. - **Rust:** Causes reddish-brown pustules on leaves, reducing photosynthetic area. - **Aflatoxin contamination:** A serious quality concern caused by Aspergillus fungus, particularly in poorly stored or moisture-damaged groundnut, which can produce harmful toxins. Using disease-resistant varieties, proper crop rotation, and correct drying and storage practices are key to managing these diseases and quality risks.

18. Deficiency Symptoms

Groundnut plants show specific symptoms when key nutrients are deficient. - **Nitrogen deficiency:** General yellowing of older leaves and stunted growth, though usually less severe than in cereals due to nitrogen fixation. - **Phosphorus deficiency:** Dark green to bluish leaves with poor root and nodule development. - **Potassium deficiency:** Yellowing and browning of leaf margins, along with poorly filled pods. - **Calcium deficiency:** Causes "pops" or poorly filled, hollow pods, since calcium is essential for kernel development inside the pod. - **Iron deficiency:** Common in alkaline soils, causing yellowing between leaf veins on young leaves (interveinal chlorosis). Addressing these deficiencies through balanced fertilization and gypsum application helps ensure healthy pod development.

19. Harvesting

Groundnut is ready for harvest when the leaves begin to turn yellow and pods show a hardened, well-netted shell with dark internal seed coat color, typically 100 to 130 days after sowing depending on variety. Harvesting is usually done by loosening the soil with a plough or digging fork and then manually pulling up the plants, since pods form underground and cannot be harvested like above-ground crops. After lifting, plants are left in the field for a few days to dry before pods are picked or stripped from the plants. Careful timing of harvest is important, since delayed harvesting can lead to pod shattering and losses in the soil, while early harvesting reduces yield and oil content.

20. Storage

After harvesting, groundnut pods should be dried thoroughly in the sun until moisture content drops to around 8 to 10 percent, which is critical to prevent fungal growth and aflatoxin contamination during storage. Pods should be stored in clean, dry, well-ventilated bags or containers, kept off the ground, and protected from moisture and pests. Regular inspection for mold, insect damage, and moisture buildup is essential, since groundnut is particularly susceptible to aflatoxin development if stored improperly. Proper drying and storage practices are especially important for groundnut intended for food use.

21. Expected Yield in Pakistan

Under rain-fed conditions in the Pothohar region, average groundnut pod yield ranges from about 10 to 15 maunds per acre (375 to 560 kilograms per acre), depending on rainfall and management practices. Under good irrigated conditions with improved varieties and proper fertilization, yields can reach 20 to 25 maunds per acre (750 to 940 kilograms per acre) or more. Actual yield depends heavily on rainfall timing, variety choice, soil fertility, and pest and disease management throughout the season.

22. Government Recommendations

Provincial agriculture research institutes, including the Barani Agricultural Research Institute (BARI) Chakwal, and the Punjab Agriculture Department, regularly issue recommendations for groundnut cultivation. These typically emphasize using certified seed of recommended varieties, applying gypsum at flowering for improved pod filling, and practicing proper post-harvest drying to reduce aflatoxin risk. Farmers are also encouraged to rotate groundnut with cereal crops like wheat to maintain soil health and break pest and disease cycles, and to consult local agricultural extension offices for updated variety and input recommendations each season.

23. Modern Farming Practices

Farmers in Pakistan's groundnut-growing areas are gradually adopting improved practices such as ridge sowing for better drainage and easier mechanized harvesting, use of Rhizobium inoculants to boost nitrogen fixation and reduce fertilizer costs, and mechanized digging equipment to reduce labor requirements and pod losses during harvest. Improved post-harvest handling, including mechanical shelling and proper drying facilities, is also becoming more common among commercial groundnut producers aiming to meet quality standards for both food and oil markets.

24. Precision Agriculture

Precision agriculture adoption in groundnut farming is still at an early stage in Pakistan but is gaining attention through research and extension programs. Emerging practices include soil testing to guide precise fertilizer and gypsum application rates, moisture sensors to optimize irrigation scheduling and avoid late-season waterlogging that can damage pods, and the use of drone imagery to monitor crop health and detect disease or pest hotspots early in larger commercial fields. These technologies are expected to become more accessible to groundnut farmers as agricultural extension services expand their outreach.

25. Sustainable Farming

Groundnut supports sustainable farming naturally because of its nitrogen-fixing ability, which reduces the need for synthetic nitrogen fertilizer and improves soil fertility for subsequent crops such as wheat. Sustainable practices encouraged for groundnut cultivation include crop rotation with cereals to break pest and disease cycles, integrated pest management to minimize chemical pesticide use, and proper residue management, where groundnut haulms (plant residues) are used as valuable livestock fodder rather than being burned. Careful water management, especially avoiding over-irrigation near maturity, also supports both crop quality and sustainable use of water resources in Pakistan's rain-fed and irrigated groundnut areas.

26. Summary Note

This guide reflects groundnut farming practices specific to Pakistan, with particular focus on the Pothohar region of Punjab along with growing areas in Khyber Pakhtunkhwa, Sindh, and Balochistan. Farmers should confirm current variety availability, gypsum and fertilizer recommendations, and sowing schedules with their local agriculture extension office, as these are updated periodically based on research findings.